

Denso AFR Wideband Controller

Testing Conditions	Testing Parameters	Result	Expected Result	Notes
Verify voltage on +Sensor and -Sensor Rails (current source unconnected)	V ₊ = 2.5V V ₋ = 2.1V	V _{+Sensor} = <u>2.575V</u> V _{-Sensor} = <u>2.083V</u> VCC = <u>11.89V</u> +5V = <u>5.146V</u> V _{diff} = <u>492mV</u>	V _{+Sensor} ≈ 2.5V V _{-Sensor} ≈ 2.1V	Differential Voltage Rails should be accurate (within <0.5%)
Verify op-amp offset (ground inputs and power with VCC = 5V and Ground)	V _{off} = ? V	V _{off} = 20.1mV	Depends on Op-amp used. MCP602x has V _{off} = 500 μV	Offset should be as small as possible.
Assembled circuit as seen in the schematic Compare results with simulation.	Typical Circuit Operation	I = 0A; λ = 2.560V I = 760μA; λ = 4.523V I = -759μA; λ = 0.690V $P_{RE} = \frac{(24mV)^2}{10\Omega}$ <u>P_{RE} = 57.6μW</u>	When I = -800μA, λ = 0V When I = 0A, λ = 2.5V When I = 800μA, λ = 5V	$P_{RE} = \frac{V_{diff}^2}{R_E}$
Assembled circuit as seen in the schematic; R _E = 1Ω Compare results with simulation and first tests.	R _E Variation	I = ____; λ = ____ I = ____; λ = ____ I = ____; λ = ____ P _{RE} = ____	When I = -800μA, λ = 0V When I = 0A, λ = 2.5V When I = 800μA, λ = 5V	$P_{RE} = \frac{V_{diff}^2}{R_E}$
Assembled circuit as seen in the schematic; R _E = 0Ω Compare results with simulation and previous tests.	R _E Removal	I = ____; λ = ____ I = ____; λ = ____ I = ____; λ = ____ P _{RE} = ____	When I = -800μA, λ = 0V When I = 0A, λ = 2.5V When I = 800μA, λ = 5V	$P_{RE} = \frac{V_{diff}^2}{R_E}$
Additional Notes or Calculations:				
Power Supply used is not great, and the chosen components aren't rail to rail, even with the workarounds, it's not perfect. 2.1V rail was adjusted with trim pot (R2 = pot) due to lack of resistors. Adjusted to 2.1V (a little higher to account for 2.578V voltage). Performance seemed alright, it calculated AFR just fine. Couldn't be bothered to change replace R_E with jumpers or resistor values. The power dissipated is very low during sink & sourcing (used highest Power value). I wouldn't lower it since at a short, the current would be $I = \frac{2.5V}{10} = 250mA.$				